

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Computer Networks & IoT

Paper ID

1984

Paper Title

Simulation of Scalable and Secure Backend Architectures for Resource-Constrained Web Applications

Abstract

Today's web applications experience the challenge of supporting the ever-increasing number of users while providing sufficient security guarantees under resource constraints like budget and computation. It is still non-trivial to determine the best backend architecture, especially for small development teams, research centers, and startups. This paper provides an empirical evaluation of the popular backend architectures, including monolithic, microservices, and hybrid, with respect to scalability and security for resource-constrained web applications. From the experimental results, the monolithic architecture is seen to perform well under low workloads, while its performance degrades considerably under high concurrency. The microservices architecture supports the maximum throughput under high concurrency, with the cost of high memory usage. The hybrid architecture strikes the right trade-off between performance and security by centralizing the security functions at the gateway layer.

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Submission Files

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Submission Questions Response**1. Corresponding author Email ID**

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